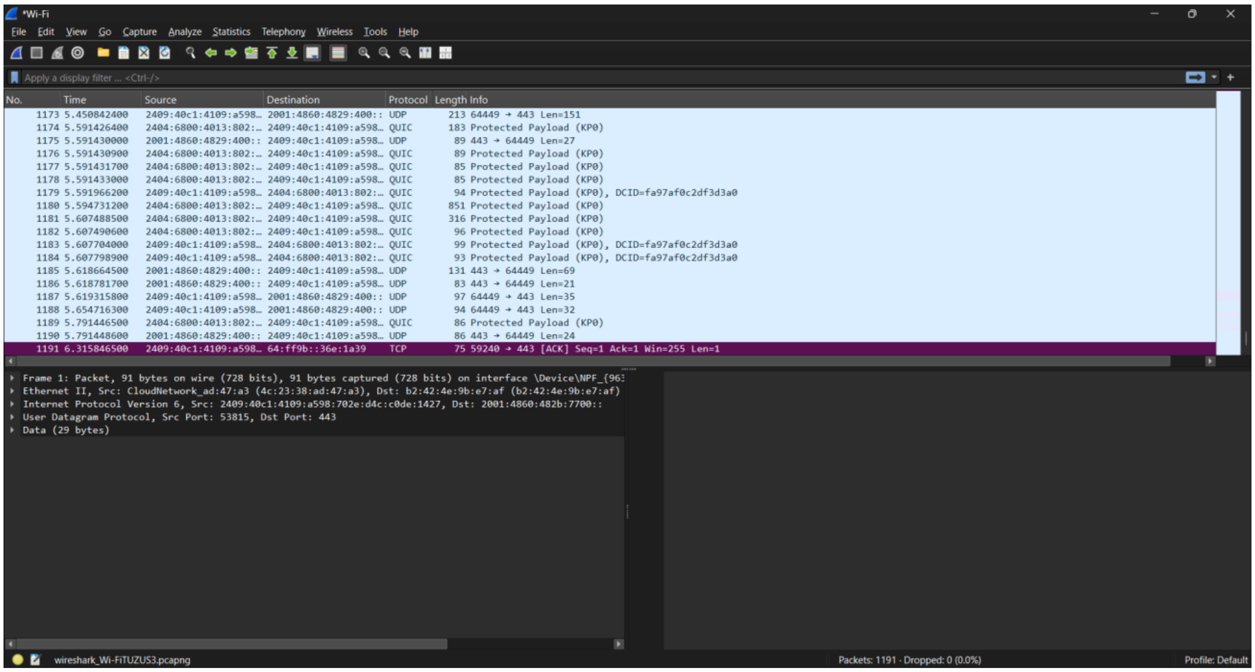
DAY-1 INTRODUCTION TO NETWORKING AND OSI MODEL

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.



Protocol	Purpose
DNS	Resolves the website name (instagram.com) to an IP address.
ARP	Finds the MAC address of devices on the local network.
TCP	Establishes a reliable connection between your device and the server.
TLSv1.3 (or TLS)	Encrypts the communication for secure browsing (HTTPS).
HTTP/2 or HTTP	Transfers web page data (usually HTTPS traffic appears over TLS).
QUIC (optional)	Used by some browsers for faster secure connections over UDP.
UDP	Used by DNS, QUIC, and other services.

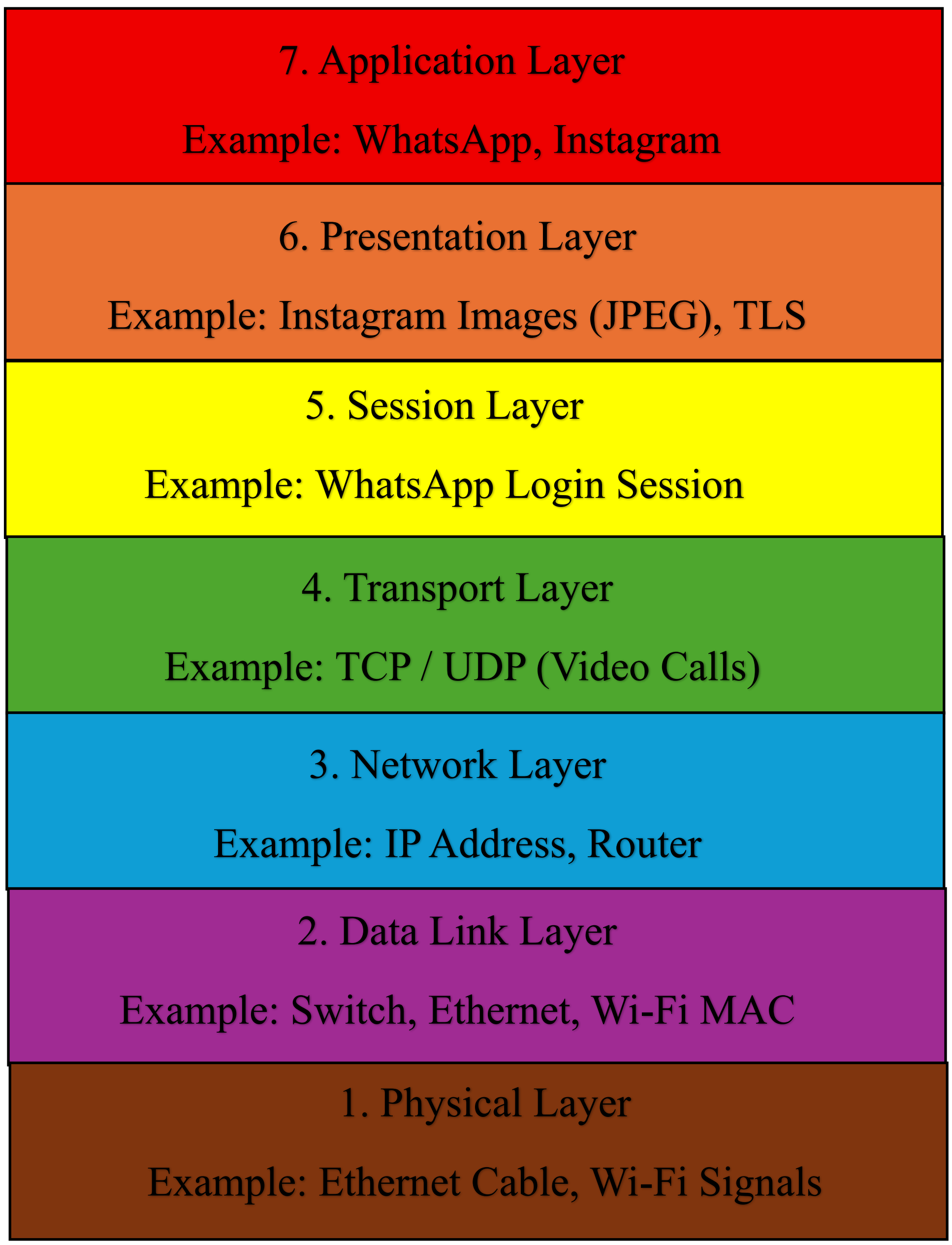
**2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

Packet (Protocol)	OSI Layer	Reason
UDP	Transport Layer (Layer 4)	Used for fast communication without guaranteed delivery (e.g., DNS, streaming, QUIC).
ARP	Data Link Layer (Layer 2)	Resolves an IP address to a MAC address for communication on the local network.
TLS	Presentation Layer (Layer 6)	Encrypts data to provide secure communication between the client and server (HTTPS).

3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

OSI MODEL



**4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.**

1. Application Layer (Layer 7)
You type a message in WhatsApp and press Send. WhatsApp prepares the message to be sent.

2. Presentation Layer (Layer 6)
The message is encrypted so that only the sender and receiver can read it. It also formats the data if needed.

3. Transport Layer (Layer 4)
The message is divided into small packets. TCP or UDP helps send the packets to the correct destination.

4. Network Layer (Layer 3)
Each packet gets the sender's and receiver's IP address. Routers use these IP addresses to send the packets over the Internet.

5. Data Link Layer (Layer 2)
The packets are converted into frames and sent using the MAC address over the local Wi-Fi or mobile network.

6. Physical Layer (Layer 1)
The data travels as electrical signals, radio waves (Wi-Fi), or mobile network signals through cables or the air to the receiver's device.